



RBHHM: A novel remote cardiac cycle detection model based on heartbeat harmonics

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ABSTRACT

Cardiac cycle detection methods based on radar sensing have made marked progress in the last decade. In this study, a novel harmonic distribution based non-contact cardiac cycle detection method is innovatively proposed to extract heartbeat intervals and rhythm sequences from radar signals. Initially, we exploit the bimodal Gaussian distribution model of the frequency domain and the Shannon energy to separate heartbeat harmonics from radar signals, which preserve instantaneous variation features of signal components. Furthermore, we cluster harmonics into three types of waves based on K-means. The peak positions of the three waves are near locations of the R-peak, T-peak, and T-wave end in the electrocardiograph (ECG) respectively. Thus, the detected peak locations of three waves can be directly used to estimate heartbeat-to-beat intervals. Additionally, we compared three recurrent neural networks (RNNs) using the clustered waves to segmentation sequences of “R-T-R” and “Q-R-S-T-Q”. Specific subsets are classified from the sequences to prove their association with cardiac systolic and diastolic variation. Finally, verification is performed using a clinical dataset. The suitable centre frequency range to use the proposed model is about 3–15 Hz. The performance of the cardiac cycle detection method achieves a mean relative error of $5.34\% \pm 3.57\%$ for BBIs detection and an accuracy of 95.88% for heartbeat rhythm segmentation. The experimental results show that our radar-based heartbeat harmonic model (RBHHM) based method has better accuracy than other state-of-the-art cardiac cycle detection methods.

1. Introduction

Currently, COVID-19 has infected many across the world, prompting people to pay more attention to their health. Typically, doctors diagnose a patient's state of health by monitoring heart activity with electrocardiograph (ECG). However, for some special patients, such as burn patients and infants, noncontact detection or remote healthcare is required. Biological radar sign detection technology has developed markedly in the past decade and has attracted much attention due to its strong anti-jamming ability and timely detection [1–8].

Cardiac movement is a complex mechanical movement caused by the regular contraction and extension of the ventricles and atria. It has been shown that the process of ventricular depolarization and repolarization can be monitored by the biological radar [9]. These two distinct processes correspond to the QRS complex and T wave processes in ECG, respectively [10,11]. The first heart sound (S1) and second heart sound (S2) are also related to the two processes of the R-peak and T-wave end

respectively [12]. In addition, the central frequency of the QRS complex wave in ECG is approximately 10–25 Hz, which is less than 40 Hz, while the centre frequency of the T-wave component is less than 10 Hz [13]. Distinct PQRST sequences of the ECG waveform can be applied in healthcare, such as heart disease diagnosis [14,15]. If different phases of the ventricular cycle can be extracted from the signals detected by radar, more reference information will be available for diagnosing heart disease compared with simple detection of noncontact heart rate or heartbeat-to-beat intervals (BBIs). However, there are still some difficulties to be solved in practical applications. The amplitude of the chest wall caused by a heart beating is smaller than that of respiratory movement, and the frequency of the second harmonic respiration is similar to the base frequency of heartbeat [16–18]. Therefore, detecting heartbeat changes by radar signals is vulnerable to interference from self-breathing and other body movements. Estimating BBIs and cardiac sequences by higher harmonics may reduce the interference of respiratory signals. Overall, effective and accurate cardiac cycle detection is

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still challenging.

The prior BBIs detection researches using biological radar include decomposition-based methods (i.e., empirical mode decomposition and wavelet decomposition) [19,20] and waveform reconstruction-based methods [21]. Therein, the sinusoidal signal is mainly used for a radar-based heartbeat motion model among those studies. Although this helps simplify the analysis of heart rate changes, the amplitude change of sinusoidal signals does not represent the amplitude change of heartbeat motion. The Gaussian pulse train model has been utilized to detect accurate heart rates [22] in the biological radar sensing, which inspires us to explore the novel heartbeat expression model based on the Gaussian model.

From the time series classification perspective, cardiac sequences can be segmented with the assistance of machine learning and deep learning. Most commonly employed features are cardiac signal envelopes, such like the homomorphic envelope (HoEnv), Hilbert envelope (HiEnv), power spectral density envelope (PSDEnv) and wavelet envelope [23], which result in accurate heart sound segmentation in the literature. In the terms of machine learning, the logistic regression-hidden semi-Markov model (LR-HSMM) has been proved to address the problem of the accurate heart sound segmentation [23]. Moreover, neural networks based deep learning methods, such as the long short-term memory (LSTM) and the gate recurrent unit (GRU), have achieved out-performances than machine learning methods for segmentation of cardiac sequences [24-26]. However, it seems that there is no research trying to segment and classify the cardiac radar sequences to by establishing the correspondence between ECG time series and cardiac radar signals.

The purpose of this research is to develop a mathematical model of cardiac radar signal to assist the cardiac cycle detection. Then effective feature information based on the proposed model is conducive to detect accurate heartbeats. Furthermore, the correlations between two cardiac sensors of ECG and radar will be studied using recurrent neural networks (RNNs). Finally, the real-word clinical dataset is utilized to evaluate the proposed mathematical model and cardiac cycle detection method. The major novelties and contributions of this research can be summarized in three aspects:

- 1) First, a novel radar based heartbeat harmonics model (RBHHM) is proposed and leads to the subsequent proposed systematic signal processing methods.
- 2) The features of heartbeat information determine the detection results of cardiac cycles. A signal pre-processing and feature extraction method is proposed based on RBHHM. The extraction method of harmonic components is developed based on bimodal Gaussian distribution model (Bi-GDM) and then the K-means based signal clustering guides three main feature waves.
- 3) The cardiac cycle detection method is proposed based on RBHHM, which is demonstrated in Fig. 1. Three RNN architectures are constructed and compared. The proposed heartbeat status detection method can annotate the remote cardiac cycles with the QRS complex and T wave of ECG signal.

The remainder of this paper is organized as follows. Section 2 introduces the construction of RBHHM as well as the RBHHM based cardiac cycle detection method. Three RNNs are constructed in this section.

The metrics to evaluate suitable RBHHM centre frequencies and verify the performance of cardiac cycle detection methods is also introduced in this section. Section 3 formulates experiments results of RBHHM and cardiac cycle detection method based on an existing open-source dataset. Section 4 discusses verification results compared with the state-of-art methods. Section 5 concludes this study.

2. Methods

2.1. Principle of vital signs acquisition using radar

The principle of biological radar detection of human physiological signals is to collect human chest wall displacement signals and separate chest wall surface vibration signals caused by breathing, heartbeat and random body movement. The ECG monitor and breathing sensor typically collect reference signals synchronously during the experiment, as shown in Fig. 2. Clinically, an ECG cycle is composed of a P wave, a QRS complex and a T wave.

For a biological radar that uses an orthogonal receiving mechanism, the output signal is multiplied by two local oscillator signals with a phase shift of 90° . After filtering the two mixed signals, the in-phase and orthogonal baseband signals $I(t)$ and $Q(t)$ can be obtained by [1]:

$$I(t) = A_I \cos\left(\frac{4\pi\Delta R(t)}{\lambda} + \theta_0 + \Delta\phi\right) \quad (1)$$

$$Q(t) = A_Q \sin\left(\frac{4\pi\Delta R(t)}{\lambda} + \theta_0 + \Delta\phi\right) \quad (2)$$

where $\Delta R(t)$ is the chest wall displacement, λ is the wavelength of the carrier wave, θ_0 is the initial phase shift, $\Delta\phi$ is the phase noise, and A_I and A_Q are the amplitudes of the same trust sign and the orthogonal signal, respectively. The extended differentiate and cross-multiply (DACM) algorithm is used to obtain the phase difference of the biological radar signal, removing the DC component and enhancing the heartbeat component [21,27,28]. The extended DACM can be expressed as:

$$\frac{d}{dt}[\varphi(t)] = \frac{d}{dt}\left[\arctan \frac{Q(t)}{I(t)}\right] = \frac{I(t)Q'(t) - I'(t)Q(t)}{I(t)^2 + Q(t)^2} \quad (3)$$

where $\varphi(t)$ is the signal phase information, and $I(t)'$ and $Q(t)'$ are the differential forms of $I(t)$ and $Q(t)$, respectively. In this research, the subsequent signal analysis is based on phase difference because of the theoretical proportional relation between the phase variation and the moved distance of the chest wall.

2.2. RBHHM: Novel radar-based heartbeat harmonics model

The heartbeat movement process detected by radar can be divided into stages associated with ventricular systole and diastole. By fitting the quadratic integral signal curve of heartbeat acceleration detected by the seismocardiography (SCG), the result is as follows [10]:

$$x(n) = g(n) \exp\left(-\frac{(n-b)^2}{c}\right) \quad (4)$$

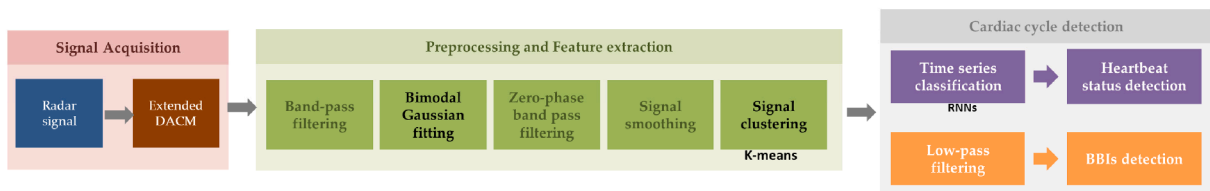


Fig. 1. The overall process of the cardiac cycle detection method based on RBHHM.

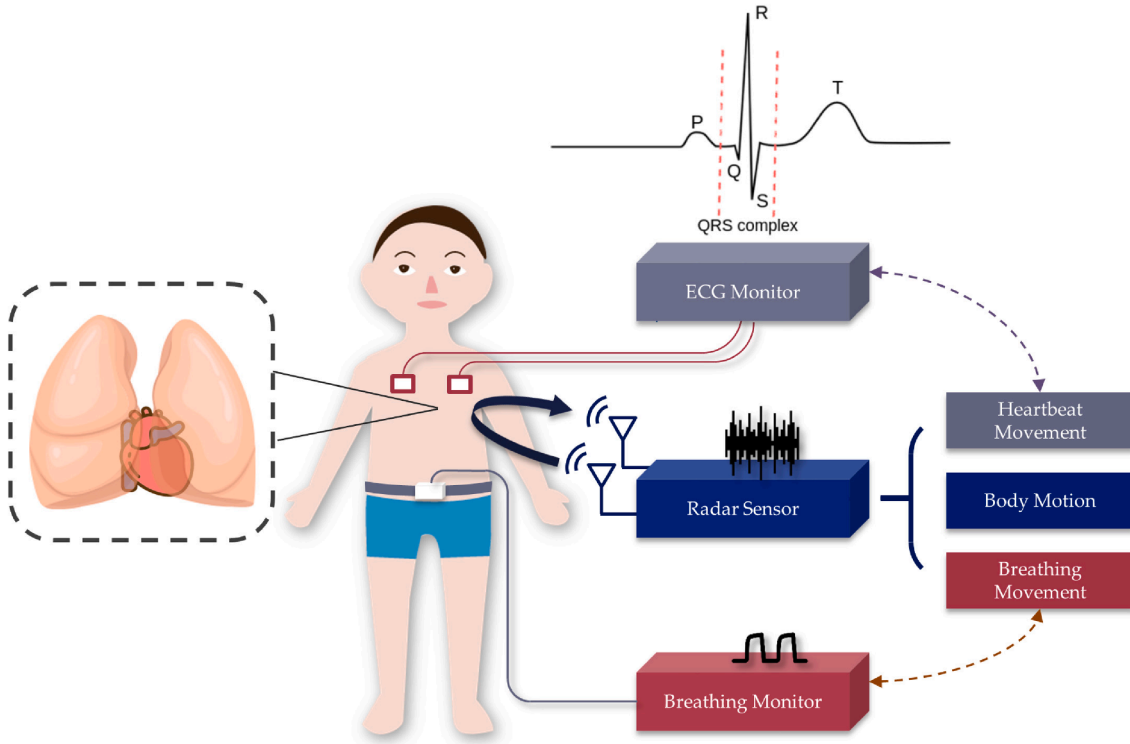


Fig. 2. Detect vital signs based on the radar compared with the ECG monitor and breathing monitor.

$$g(n) = \eta \cos(\omega n + \gamma \sin(\Omega n)) \quad (5)$$

where n is the number of sampling points; b and c are constants; and $g(n)$ is the conditional equation obtained by fitting critical data. The quantities ω and Ω are the conditioning parameters to determine the location of the peaks.

The time-domain diagram and unilateral spectrum analysis diagram of a signal from 3 to 50 Hz are shown in Fig. 3(a) and (b), demonstrating that the respiratory signal is suppressed sufficiently in the higher frequency band, while the harmonic component of the heartbeat is primarily retained. Considering that previous studies [11,29] have estimated the heart rate from the high-frequency part of the radar signal, RBHBM is proposed in combination with Eq. (4) and the frequency distribution of each wave component in the ECG:

$$x_h(t) = \exp\left(-\frac{(t-b)^2}{c}\right) \sum_{k=K_{min}}^{K_{max}} a_k \cos(2\pi k f_0 t + \theta_k) \quad (6)$$

where f_0 is the heart rate; k is the harmonic order of heartbeat signal; K_{min} and K_{max} the minimum number and the maximum number of harmonic orders within the frequency range of 3–40 Hz, respectively; and a_k and θ_k are the amplitude and phase shift of the corresponding harmonics component, respectively. This formula results from mutual modulation between the Gaussian pulse signal of the radar and the harmonic part of the heartbeat signal. The Shannon energy describes the instantaneous variation of the signal and is beneficial to detect peak location, and can be computed as follows:

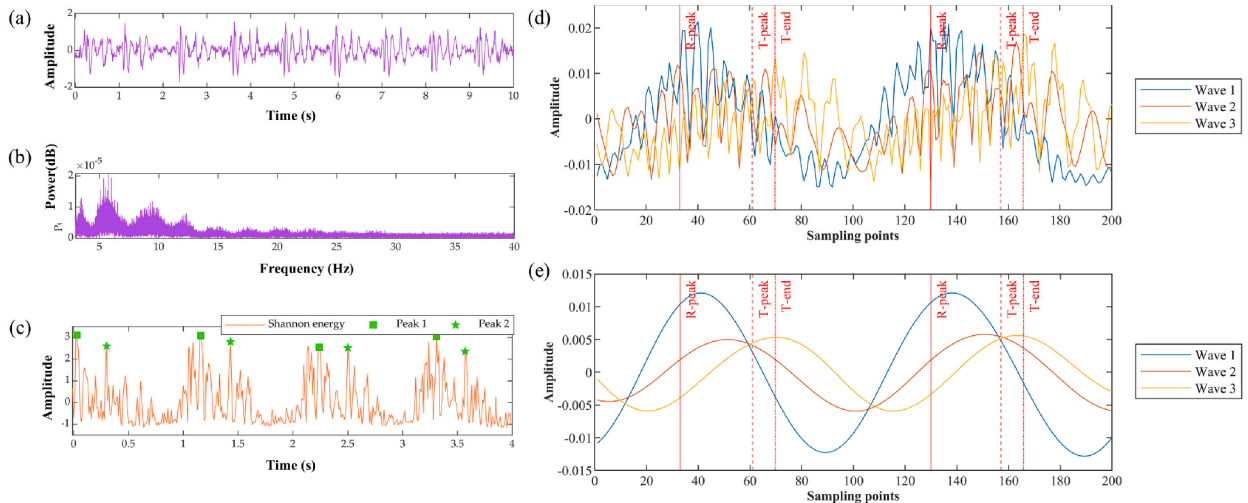


Fig. 3. Radar vital signal after filtering out the respiratory components. (a) Variation in the time domain. (b) Unilateral spectrum analysis result. (c) Shannon energy envelopes of radar signals in the time domain. (d) The clustering results of three waves. (e) Low-pass filtered clustering results.

$$x_n(t) = \frac{x(t)}{\max(|x|)} \quad (7)$$

$$Se(t) = -x_n^2(t) \log(x_n^2(t)) \quad (8)$$

where $x_n(t)$ is the normalized signal and $Se(t)$ is the Shannon energy. The curve of Shannon energy in Fig. 3(c) shows that, compared with the amplitudes of components involving Peaks 1 and 2, the amplitudes of other harmonic components and interference components are much smaller. Therefore, the magnitude of the components except Peaks 1 and 2 is ignored. Thus, the RBHMM can be further simplified as:

$$x_h(t) = [A_1 \cos(2\pi f_{k_1} t + \varphi_1) + A_2 \cos(2\pi f_{k_2} t + \varphi_2)] \exp\left(-\frac{(t-b)^2}{c}\right) \quad (9)$$

where A_1 and A_2 are the harmonic amplitudes of Peak 1 and Peak 2, respectively; k_1 and k_2 are the corresponding harmonic times; and φ_1 and φ_2 are the corresponding phase shifts. The discrete expression after sampling is as follows:

$$x_h(n) = \sum_l A_l \cos(\omega_l n + \varphi_l) \exp\left(-\frac{(n-b)^2}{c}\right) \quad (10)$$

where l is equal to 1 or 2 based on the component number of Peak 1 and Peak 2. Fourier transform of Eq. (10) can be described by follows:

$$X(\omega) = \sum_l v_l \times [\delta(\omega - \omega_l) + \delta(\omega + \omega_l)]^* \sqrt{\frac{\pi c}{2}} \times \exp(-j\omega b) \times \exp\left(-\frac{\omega^2}{4}\right) \quad (11)$$

where $v_l = A_l e^{j\varphi_l} \pi/2$. It is known that the Fourier transform result of the Gaussian pulse presents a Gaussian distribution with a frequency of 0 Hz at the centre of symmetry. When the Gaussian pulse signal is multiplied by the cosine signal of a constant frequency ω_c in the time domain, the Gaussian distribution's central frequency is shifted to the frequency domain. After normalization of the unilateral spectrum analysis data of the sampled signal, the data can be approximately fitted into a bimodal Gaussian distribution model (Bi-GDM) with ω_1 and ω_2 as the centre frequency of symmetry:

$$|X(\omega)| \leftarrow \sum_l \phi_l \frac{1}{\sqrt{2\sigma_l^2 \pi}} \exp\left(-\frac{(\omega - \omega_l)^2}{2\sigma_l^2}\right) \quad (12)$$

where ϕ_l is the weight coefficient, and σ_l^2 is the distribution variance. Therefore, after the heartbeat signal within 3–40 Hz is separated from the radar signal, two centre frequencies of the signal can be obtained by fitting Bi-GDM in the frequency domain introduced during pre-processing. The purpose waveforms of centre frequencies can be obtained using band-pass filtering. The possible centre frequencies may have a broad range, but filtered waveforms can be clustered into three categories using the Pearson correlation coefficient as the clustering distance, as demonstrated in Fig. 3(d). Then, using a low-pass filter of the heartbeat frequency range, three waves are smoothed, and peak locations are more marked in Fig. 3(e). Compared with synchronously sampled ECG, the peaks of 3 clustered waves are associated with the R-peak, T-peak and T-end, where the locations of T-end are shifted from T-peaks with 88 ms [30].

For the ECG waveform's distinctive PQRS components, the original dynamical equations of heartbeat motion in Cartesian coordinates are given as follows [31]:

$$\dot{x} = \gamma x - \omega y \quad (13)$$

$$\dot{y} = \gamma y - \omega x \quad (14)$$

$$\dot{z} = - \sum_{i \in \{P, Q, R, S, T\}} \alpha_i \Delta \theta \exp\left(-\frac{\Delta \theta_i^2}{2b_i^2}\right) - (z - z_0) \quad (15)$$

where $\alpha = 1 - \sqrt{x^2 + y^2}$, $\Delta \theta_i = (\theta - \theta_i) \bmod 2\pi$, $\theta = \text{atan}(y, x)$. Compared with Eqs. (9) and (12), the ECG waveform (15) is also the sum of distinct exponential equations. Therefore, if the P-wave in the ECG is ignored, the segments of the QRS-wave and T-wave have the same dominant proportion and amplitude characteristics.

2.3. Pre-processing and feature extraction

Information extracted from vital radar signs contains heartbeat, respiratory and other interferences. Therefore, the signals in the 3–40 Hz range must be extracted first. After the extended DACM algorithm acquires the radar phase signal, the respiratory interference and high-frequency noise can be filtered via zero-phase band-pass filtering, retaining the phase shift of the original signal as much as possible. To acquire more accurate heartbeat information, a pre-processing method of the cardiac cycle detection method is proposed as follows:

Analyse the unilateral spectrum and extract the centre frequency. Two centre frequencies are obtained using Bi-GDM fitting of the frequency band divided by the sliding window. The centre frequencies whose fair values are too close will be replaced by their mean values. The specific frequency window limits will be analysed in Section 3.1.

Estimate heartbeat harmonic components. Zero-phase band-pass filtering with cut-off frequencies, using estimated values as centre frequencies, is used to obtain estimated heartbeat harmonic components with a constant frequency bandwidth.

Each signal component is smoothed separately. Each signal is normalized by dividing by the maximum absolute value at the beginning. The Shannon energy of the output signals after normalization is calculated. Then, zero-phase filtering using a rectangular impulse response is performed. The length of the rectangular window is 0.4 s. Finally, the baseline drift is eliminated by subtracting the smoothed signal with a window of 1.5 s. The specific theoretical process in the fourth step refers to [32].

Cluster all signals into three categories based on K-means. The cluster distance is the Pearson correlation coefficient between two signals. Clustered centre waves correspond to variation limits of the T-end, R-peak, and T-peak.

The following problems may be encountered when selecting the frequency band to fit Bi-GDM: (1) the primary centre frequencies of the two components are unknown; (2) the difference between the centre frequencies of the two components is too small to be determined by the fitting; and (3) the separated signal may be the harmonic signal or superposition signal of the two components. These questions can be solved by the proposed pre-processing and detection method, which will be verified in Section 3.

2.4. Cardiac cycle detection

BBIs and heartbeat status are detected using the extracted three waves and compared with the cardiac cycle results of ECG signals. The heartbeat intervals of ECG can be computed by the intervals of R-peak to R-peak (RR) or T-peak to T-peak (TT). In addition, there are two annotation methods of "R-T-R" and "Q-R-S-T-Q" using ECG sequences as shown in Fig. 4. Labels of cardiac activation status including RT (labelled as "1") and TR (labelled as "2") and the durations are extracted from the locations of the R-peak and T-peak. Similarly, the positions of Q, R, S, and T detected from ECG indicate different ventricular diastole and contraction status and lead to the four sequences QR (labelled as "1"), RS (labelled as "2"), ST (labelled as "3"), and TQ (labelled as "4") respectively.

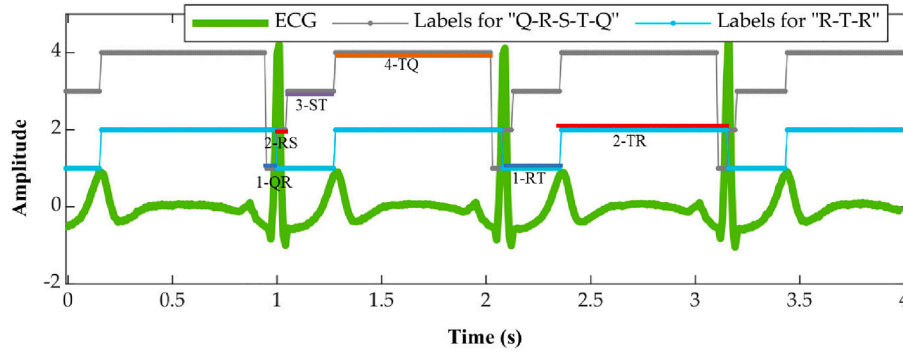


Fig. 4. The annotations of cardiac activation status using ECG sequences.

2.4.1. Detection of heartbeat to beat intervals

The heartbeat locations can be detected by low-pass filtering and peak searching to compute BBIs. An 8th-order Butterworth low-pass filtering with a cut-off frequency of 1.5 Hz is used to eliminate redundant peaks and noise. Because a heartbeat cycle is about 0.5 to 2 s, local peaks with a heartbeat cycle spacing are detected by searching for the local maximum point where the different values of the two sides have opposite signs. Therefore, the time-varying BBIs can be calculated by the forwards difference of the peak time positions.

2.4.2. Detection of heartbeat sequences

The three clustered waves can be used to segment the cardiac cycle into micro sequences based on RNNs. As shown in Fig. 5, different RNNs models are constructed. Fig. 5(a) demonstrates the architecture of the bidirectional LSTM (Bi-LSTM) based bidirectional RNN (Bi-RNN). Fig. 5 (b) shows the architecture of deep RNN (DRNN) using two layers of LSTM or GRU networks. Specifically, the first layer embeds the vector with the clustered features. The size of the input vector is 3 depending on the number of feature waves. For the employed Bi-RNN, there are 200 nodes in the Bi-LSTM hidden layer. For the employed DRNN, there are 100 nodes in the first hidden LSTM (or GRU) layer and 50 nodes in the second LSTM (or GRU) hidden layer. Sequentially, a fully connected layer connects the output nodes of extracted latent time-series information and a softmax layer. The output layer is the category of cardiac activation status with the most probability from the softmax layer.

The RNN models are trained by the adaptive momentum estimation. The learning rate is initially set to 0.01 and updated for every two epochs. There are 30 training iterations for all models. For each subject, the proportion of the training set compared to the test set is 7:3. The tolerance window is 10 ms according to the sampling frequency.

2.5. Dataset and evaluate metric

The proposed RBHMM and cardiac cycle detection method were

evaluated using the clinically recorded open-source dataset [33]. This dataset contained 24 h of synchronized data from a 24-GHz CW radar and a reference device. Thirty healthy subjects were measured in five scenarios: “Resting”, “Valsalva”, “Apnoea”, “Tilt-Up” and “Tilt-Down”. In most clinical application scenarios, people are in a state of relaxation and rest. Thus, the dataset for the rest scenario is used.

To verify the correlation between the two signal models and find the best centre frequency range suitable for detecting Peaks 1 and 2, all possible centre frequencies are analysed. Locations of R-peaks (Rlocs) and areas of T-peaks (Tlocs) are detected to calculate two types of BBIs [34]. Similarly, peak locations (Plocs) of each smoothed signal (the output of the third step in the cardiac cycle detection method) are also detected to calculate BBIs. To distinguish different BBIs, RR and TT represent detected heartbeat intervals of ECG, and PP is the intervals detected from the low-pass filtered radar signal. Due to the broad range of possible centre frequencies, peaks detected from the filtered signal include peak 1, peak 2 and nearby interference peaks. The Pearson correlation coefficients $r(RR, PP)$ and $r(TT, PP)$ are used for quantitative validation for each smoothed signal, and related equations are shown as follows:

$$r(RR, PP) = \frac{\sum_{i=1}^n (RR_i - \overline{RR})(PP_i - \overline{PP})}{\sqrt{\sum_{i=1}^n (RR_i - \overline{RR})^2} \sqrt{\sum_{i=1}^n (PP_i - \overline{PP})^2}} \quad (16)$$

$$r(TT, PP) = \frac{\sum_{i=1}^n (TT_i - \overline{TT})(PP_i - \overline{PP})}{\sqrt{\sum_{i=1}^n (TT_i - \overline{TT})^2} \sqrt{\sum_{i=1}^n (PP_i - \overline{PP})^2}} \quad (17)$$

where n is the number of intervals, and RR_i , PP_i and TT_i refer to the i th heartbeat interval. To analyse the possible frequency range, all outputs of the third step in the cardiac cycle detection method are filtered, and no clustering operation is performed. Then, BBIs are extracted by peak detection from all filtered waveforms. RR, TT, and PP are interpolated into the same length. The outliers of heartbeat intervals are filled by

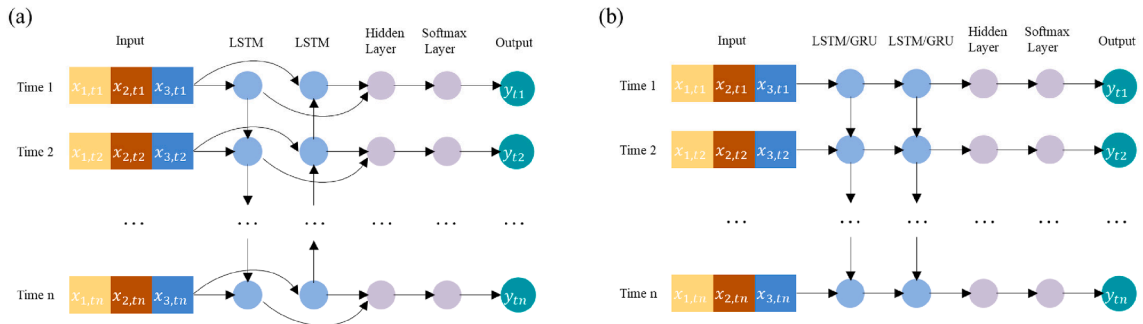


Fig. 5. The RNN architectures used in this research. (a) The architecture of Bi-RNN using a Bi-LSTM. (b) The architecture of DRNN using the two-layer LSTM or two-layer GRU networks.

linear interpolation of adjacent non-outliers.

The root mean square error (RMSE) is also used to evaluate possible centre frequencies of RBHMM. R_i , T_i and P_i mean the i th detected heartbeat location, respectively. $\text{RMSE}(\text{RR}, \text{PP})$, $\text{RMSE}(\text{TT}, \text{PP})$, $\text{RMSE}(\text{Rlocs}, \text{Plocs})$ and $\text{RMSE}(\text{Tlocs}, \text{Plocs})$ can be computed by using the following equations, respectively:

$$\text{RMSE}(\text{RR}, \text{PP}) = \sqrt{\frac{\sum_{i=1}^n (RR_i - PP_i)^2}{n}} \quad (18)$$

$$\text{RMSE}(\text{TT}, \text{PP}) = \sqrt{\frac{\sum_{i=1}^n (TT_i - PP_i)^2}{n}} \quad (19)$$

$$\text{RMSE}(\text{Rlocs}, \text{Plocs}) = \sqrt{\frac{\sum_{i=1}^{n+1} (R_i - P_i)^2}{n+1}} \quad (20)$$

$$\text{RMSE}(\text{Tlocs}, \text{Plocs}) = \sqrt{\frac{\sum_{i=1}^{n+1} (T_i - P_i)^2}{n+1}} \quad (21)$$

The mean relative error (MRE) is computed to validate the performance of BBIs detection as follows:

$$\text{MRE}(BBI_{\text{radar}}, BBI_{\text{ECG}}) = \frac{1}{n} \sum_{i=1}^n \frac{|BBI_{\text{radar}}(i) - BBI_{\text{ECG}}(i)|}{BBI_{\text{ECG}}(i)} \times 100\% \quad (22)$$

where BBI_{radar} and BBI_{ECG} are the heartbeat intervals detected from radar and ECG, respectively. Similarly, BBI_{radar} includes p_1p_1 , p_2p_2 and p_3p_3 , while BBI_{ECG} includes RR and TT.

The confusion matrix is counted for each subject. True positive (TP) indicates that the estimated position is classified into the correct category, false-negative (FN) indicates the estimated position is classified into the wrong category, and false positive (FP) describes categories that were wrongly assigned to other estimated positions. The following equations compute the precision, recall, and F_1 scores:

$$p = \frac{TP}{TP + FP} \times 100\% \quad (23)$$

$$r = \frac{TP}{TP + FN} \times 100\% \quad (24)$$

$$F_1 = 2 \frac{pr}{p + r} \quad (25)$$

where p is the precision, and r is the sensitivity. In addition, the proportions that are correctly classified are also computed as accuracy (Acc.).

3. Results

3.1. Evaluation of RBHMM centre frequencies

The most suitable centre frequencies to fitting Bi-GDM and RBHMM are assessed using correlation verification and root mean square error (RMSE). The statistical boxplot of centre frequencies with strong cor-

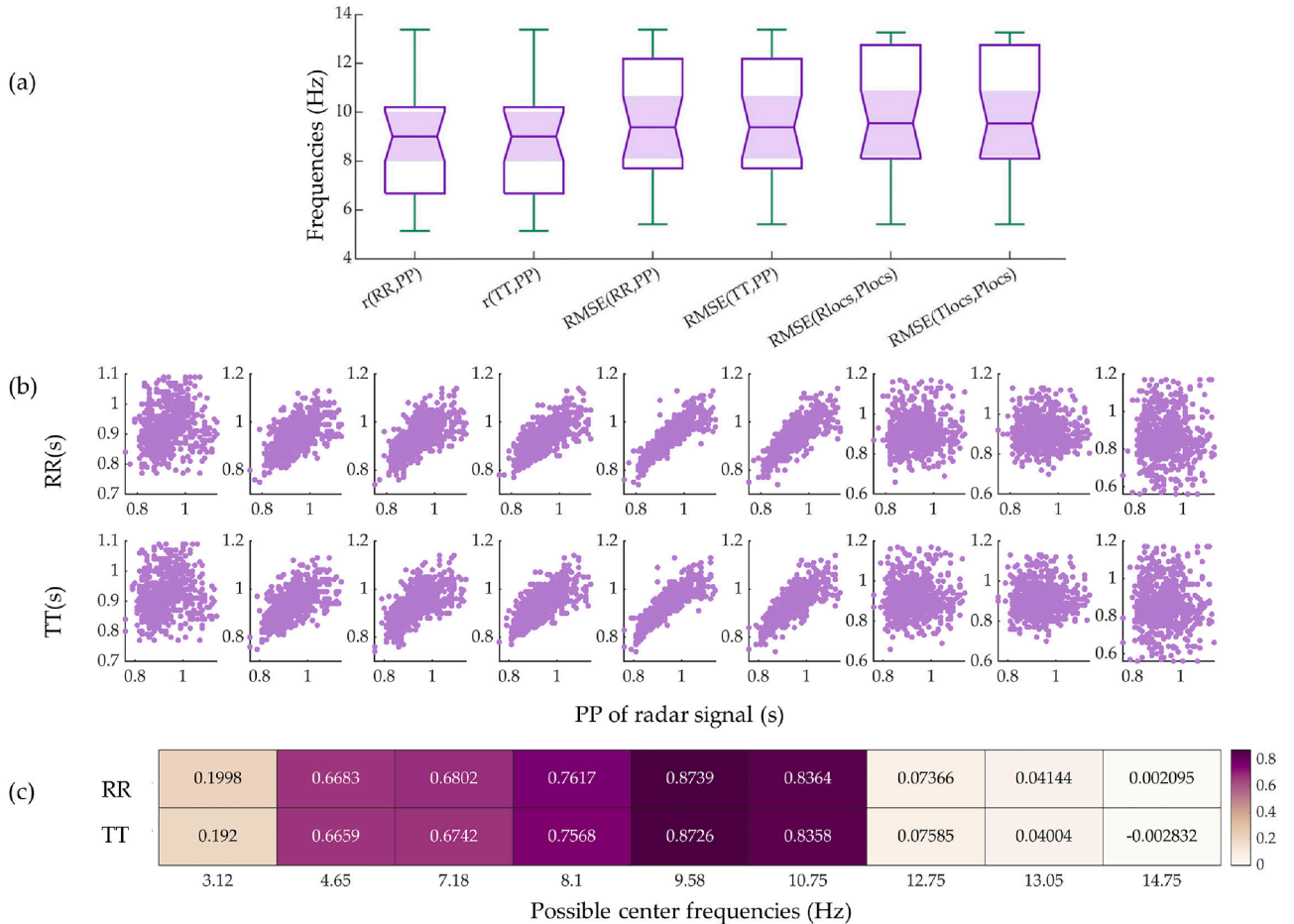


Fig. 6. Correlation verification between RR(TT) and PP of radar signals.

relation and low error values is shown in Fig. 6(a). Most centre frequencies are between 3 and 15 Hz, and the statistical median is between 8 and 10 Hz. The truncation range of zero-phase pass-band filtering with all centre frequencies at ω_{center} is $(\omega_{center} - 1.2, \omega_{center} + 1.2)$ Hz to unify the bandwidth of all target centre frequencies. For the specific centre frequencies evaluation of Subject 15, the consistency analysis of centre frequencies between 3 and 15 Hz are formulated in Fig. 6(b) and (c). The scatter charts are shown in Fig. 6(b), and the correlation map is Fig. 6(c). Nine possible centre frequencies are detected after using the Bi-GDM to fit the frequency distribution and eliminate the close values. The frequency for the maximum correlation coefficient is 9.58 Hz, where detected PP has good consistency with RR and TT. The maximum Pearson correlation coefficient $r(RR, PP)$ and $r(TT, PP)$ are 0.8739 and 0.8726 at 9.58 Hz, and the second is at 10.75 Hz. Thus, the most suitable centre frequencies range for Subject 15 to fit RBHBM is 9–11 Hz, and also corresponds to the 7th to 14th harmonic frequencies of the heartbeat.

3.2. Evaluation of cardiac cycle detection on the dataset

3.2.1. Performance of BBIs detection

The locations of the R-peak and T-peak extracted from synchronous ECG are defined as heartbeat valuation references. According to the rough estimates of the optimal centre frequency, the frequency band of emphasis in verifying the heartbeat cycle sequence detection method is 3–15 Hz. The K-means algorithm that uses the interclass correlation coefficient as the distance is used for clustering, and the centroid of each class is the clustering output signal and arranged in descending order according to the statistical variance of the signals.

The peak positions of the centroid signals of the three types of clusters correspond to the R-peak (Wave 2), T-peak (Wave 3), and T-end (Wave 1). ECG signals are collected simultaneously. Thus, three types of

peaks are detected from radar signals: p_1 , p_2 and p_3 . The results of MRE and total mean error are shown in Table 1. The mean MRE of 30 subjects between radar peak difference values (p_1p_1 , p_2p_2 and p_3p_3) and ECG heartbeat to beat intervals (RR and TT) is from 4.33% to 6.70%. Particularly, the subject 6 has the minimum MRE of RBHBM based BBIs detection results, and has a minimum MRE(p_2p_2 , TT) of 1.4%. The total MRE for 30 subjects is $5.34\% \pm 3.57\%$. For the stability evaluation of the RBHBM based BBIs detection algorithm, the standard deviation (Std.) of MRE presents a range from 3.31% to 5.77%.

3.2.2. Performance of heartbeat status detection

To detect more specific cardiac information, the sequences are segmented and classified using RBHBM and Bi-LSTM. The assessment results for the test set are shown in Table 2. The results of Subjects 21 and 22 show extreme disparities that originate from abnormal ECG signals in which the T-peak cannot be detected correctly. Thus, the mean and standard deviation are calculated without consideration of Subjects 21 and 22. Due to the low quality of the radar signal, the F_1 scores of Subjects 5, 8, and 17 are lower than those of other subjects. Without considering the subjects' data that are indicative of poor system performance, the results of the remaining cases are compared with the initial results [9] listed in the last column.

For "R-T-R" sequences, the RBHBM based Bi-LSTM training results have a good performance in F_1 scores, which represent high precision p and high recall r . Especially for subjects with good signal quality, their F_1 scores are greater than 90%, and even mostly come to 95%–99%. The F_1 -score standard deviation values of RT and TR are 6.53% and 2.15% respectively, and the low deviation expressed the preferable stability of RBHBM based algorithm. For the more subdivided sequences "Q-R-S-T-Q", the F_1 scores of TQ are better than QR, RS and ST, on account of longer time intervals of TQ. The second F_1 score is for ST. For the statistical F_1 -score results of all subjects, the low standard deviations correspond to high mean values. Therefore, the RBHBM based cardiac sequences segmentation approach has more stability in TQ annotation.

Table 1

Validation results of MRE (%) for RBHBM based BBIs detection.

Subject	MRE (p_1p_1 , RR)	MRE (p_1p_1 , TT)	MRE (p_2p_2 , RR)	MRE (p_2p_2 , TT)	MRE (p_3p_3 , RR)	MRE (p_3p_3 , TT)	MRE, AVG
1	4.02	4.20	3.50	3.65	2.66	3.00	3.51
2	4.07	4.55	3.88	4.23	17.4	17.34	8.58
3	5.35	6.06	4.55	5.38	3.87	4.89	5.02
4	5.03	5.09	2.22	2.56	2.73	3.03	3.44
5	6.23	6.42	5.10	5.21	3.01	3.17	4.86
6	2.57	2.73	1.42	1.4	1.42	1.54	1.85
7	18.18	18.37	7.34	8.09	9.22	9.94	11.86
8	11.99	12.06	7.09	7.18	5.05	5.35	8.12
9	11.97	12.37	11.93	12.37	12.02	12.52	12.2
10	3.56	3.43	3.36	3.19	3.27	3.16	3.33
11	5.14	5.53	7.14	7.56	5.14	5.54	6.01
12	2.84	2.76	5.16	5.62	3.75	4.11	4.04
13	3.35	3.57	3.13	3.41	2.32	2.52	3.05
14	28.83	28.88	24.34	24.45	3.54	3.62	18.94
15	2.41	2.94	2.63	2.23	2.32	2.51	2.51
16	5.12	5.44	4.24	4.54	4.8	5.17	4.89
17	13.54	13.58	2.50	2.54	2.11	2.16	6.07
18	3.80	3.87	5.65	5.65	2.46	2.66	4.01
19	3.31	3.32	2.70	2.77	2.80	2.85	2.96
20	4.84	5.95	3.95	4.83	3.97	5.27	4.80
21	3.00	3.36	2.59	2.91	2.71	3.13	2.95
22	12.63	12.46	2.47	2.6	2.99	2.94	6.02
23	3.91	4.48	3.64	4.25	4.15	4.45	4.15
24	4.67	4.50	4.52	4.08	2.60	3.03	3.90
25	2.03	2.26	4.26	4.44	2.33	2.75	3.01
26	5.71	6.33	5.77	5.71	2.99	3.16	4.95
27	2.01	2.30	2.12	2.48	2.69	3.03	2.44
28	3.91	4.41	3.51	3.75	4.21	4.88	4.11
29	3.29	3.89	2.37	2.94	3.72	3.54	3.29
30	5.59	5.87	5.67	6.10	7.53	8.23	6.50
mean	6.43	6.70	4.96	5.20	4.33	4.65	5.34
Std.	5.77	5.71	4.23	4.25	3.31	3.36	3.57

3.3. Comparison of different features

As shown in Table 3, different results of BBIs detection using envelopes, principal component analysis (PCA) and k-means are compared. The envelopes including HoEnv, HiEnv and PSDEnv are compared with clustered three waves for the BBIs detection. The envelopes are extracted from extend DACM signal. Then a band-pass filter of 0.75–1.75 Hz is utilized to smooth the envelopes. The peaks are detected and peak intervals are computed from three envelopes. The HiEnv has lower MRE of 8.61%–8.65% compared HoEnv and PSDEnv in the BBIs detection. PCA based BBIs detection method is also made a comparison with K-means by extracting the three main components. The first component wave of PCA leads to the highest amplitude and the lower error of BBIs compared with the second and third components. It seems that the PCA extracted waves have weaker consistence with the peaks of ECG except for the first component. In contrast to the PCA based main waves, the K-means based clustered waves can exploit the cardiac cycle information and make the equally significant among extracted waves.

3.4. Comparison of different models

Table 4 illustrates the comparison results for the cardiac status classification using different input features and different RNN architectures. In the terms of features, the envelope features include HoEnv, HiEnv, PSDEnv and wavelet envelope. The PCA based feature are extracted after Bi-GDM fitting and band-pass filtering in 0.75–1.75 Hz. The tolerance window is 10 ms according to the mentioned sampling frequency of 100 Hz. It is obvious that the Bi-GDM fitting and K-means based input feature outperforms other two features. Especially in the sequences of QR and RS, K-means based features achieve 76.78% F_1 -score and 78.08% F_1 -score respectively. However, the detection results

Table 2F₁ scores (%) and accuracy (%) for the cardiac sequence detection method based on RBHMM and Bi-LSTM.

Subject	F _{1, RT}	F _{1, TR}	Acc.	F _{1, QR}	F _{1, RS}	F _{1, ST}	F _{1, TQ}	F _{1, AVG}	Acc.
1	93.72	96.43	95.45	80.89	81.23	94.43	96.45	88.25	94.21
2	91.68	96.67	95.24	79.96	80.28	92.15	96.78	87.29	93.94
3	94.27	97.89	96.91	80.09	81.97	93.14	97.64	88.21	95.07
4	95.87	98.11	97.41	86.67	87.16	95.82	98.07	91.93	96.26
5	86.32	94.46	92.12	56.67	56.85	84.31	94.61	73.11	88.18
6	97.66	99.13	98.73	87.04	87.18	95.08	98.69	92.00	96.62
7	87.29	94.88	92.7	67.81	74.37	87.84	94.58	81.15	90.87
8	67.8	90.99	85.91	28.65	23.14	67.07	89.81	52.17	81.33
9	97.23	99.07	98.6	90.00	86.68	96.56	98.97	93.05	97.50
10	98.31	99.52	99.25	93.63	93.82	98.13	99.48	96.27	98.83
11	94.26	97.9	96.93	85.58	87.08	93.47	97.72	90.96	95.65
12	86.14	94.87	92.52	51.88	56.31	85.36	94.74	72.07	89.03
13	89.78	95.16	93.43	65.25	66.85	88.66	94.34	78.77	90.19
14	87.79	94.45	92.37	61.45	60.82	81.72	92.73	74.18	86.18
15	95.91	98.26	97.55	82.6	86.00	95.85	97.91	90.59	96.13
16	97.38	99.16	98.73	88.33	88.86	96.98	99.08	93.31	97.83
17	82.85	93.79	90.88	45.09	47.3	75.54	93.53	65.37	85.51
18	87.96	95.27	93.21	65.63	72.88	86.46	94.82	79.95	90.45
19	97.11	99.01	98.52	85.31	87.74	95.58	98.89	91.88	97.16
20	96.17	98.42	97.76	84.08	87.28	96.03	98.13	91.38	96.42
21	99.46	0	98.92	89.28	90.46	98.8	00.00	69.63	97.41
22	99.00	0	98.01	61.66	55.61	95.56	00.00	53.21	91.68
23	96.55	98.86	98.29	85.01	82.35	96.15	98.78	90.57	96.90
24	97.77	99.01	98.63	92.64	94.17	97.21	98.79	95.70	97.82
25	95.24	98.01	97.19	88.94	91.57	94.21	97.85	93.14	96.10
26	96.14	98.5	97.84	87.4	91.15	96.17	98.52	93.31	97.17
27	97.04	98.99	98.49	88.89	90.98	96.78	98.95	93.90	97.66
28	95.90	98.63	97.94	84.84	86.24	95.52	98.41	91.25	96.75
29	95.68	98.39	97.65	84.09	86.12	95.5	98.21	90.98	96.45
30	88.93	96.21	94.36	71.29	69.73	87.84	95.65	81.13	91.95
mean	92.46	97.14	95.88	76.78	78.08	91.41	96.86	85.78	93.86
Std.	6.53	2.15	3.19	15.94	16.45	7.31	2.39	10.43	4.49

Table 3

Comparison results among envelopes, PCA and K-means.

Method	MRE (p_1p_1 , RR)	MRE (p_1p_1 , TT)	MRE (p_2p_2 , RR)	MRE (p_2p_2 , TT)	MRE (p_3p_3 , RR)	MRE (p_3p_3 , TT)
Envelopes [24]	8.95	8.99	8.61	8.65	9.59	9.63
PCA	5.74	5.75	8.19	8.20	14.40	14.41
K-means	6.43	6.70	4.96	5.20	4.33	4.65

in the QR and RS based on envelopes and PCA are lower 60%. Additionally, the Bi-GDM based harmonic components extraction methods can reserve more cardiac status information relating to the QR and RS compared with envelope features. Furthermore, the constructed Bi-RNN yields the higher F₁-score in the sequence of QR and RS compared with LSTM and GRU based DRNN. The LSTM based DRNN achieves the greatest F₁-score in the longer sequences of ST and TQ. Generally, the Bi-LSTM based segmentation model with the feature from Bi-GDM fitting and K-means clustering make the greatest F₁-score and accuracy in the cardiac sequences detection.

Table 4

Comparison of with different input features and different RNN architectures. The tolerance window is 10 ms.

Model	Features	F _{1, QR}	F _{1, RS}	F _{1, ST}	F _{1, TQ}	F _{1, AVG}	Acc.
DRNN(GRU)	Envelopes [23]	35.12 ± 21.71	45.23 ± 21.40	81.40 ± 14.26	91.40 ± 4.32	61.77 ± 14.72	85.69 ± 6.25
	Bi-GDM + PCA	56.58 ± 12.76	50.39 ± 15.19	89.03 ± 4.58	95.70 ± 1.61	71.33 ± 9.72	90.21 ± 2.93
	Bi-GDM + K-means	66.63 ± 18.83	65.25 ± 16.86	90.45 ± 7.43	92.24 ± 2.76	77.88 ± 12.44	91.63 ± 4.90
DRNN(LSTM)	Envelopes [23]	40.24 ± 20.87	50.72 ± 19.88	87.34 ± 10.18	93.75 ± 3.37	66.45 ± 13.17	88.49 ± 5.19
	Bi-GDM + PCA	59.93 ± 13.57	55.98 ± 14.49	90.51 ± 3.98	96.13 ± 1.45	74.04 ± 9.83	91.16 ± 2.97
	Bi-GDM + K-means	70.72 ± 15.67	69.04 ± 15.58	93.11 ± 5.42	96.86 ± 1.78	80.82 ± 9.51	93.21 ± 3.89
Bi-RNN (Bi-LSTM)	Envelopes [23]	27.20 ± 16.10	35.06 ± 18.16	66.30 ± 20.57	86.57 ± 4.10	52.27 ± 12.33	71.39 ± 20.54
	Bi-GDM + PCA	45.61 ± 14.67	36.02 ± 19.11	86.68 ± 5.24	95.34 ± 1.74	64.29 ± 10.51	88.52 ± 3.22
	Bi-GDM + K-means	76.78 ± 15.94	78.08 ± 16.45	91.41 ± 7.31	96.86 ± 2.39	85.78 ± 10.43	93.86 ± 4.49

4. Discussion

The BBIs extracted from clustering waves do not exhibit many differences compared to the reference signals. Thus, the extracted signals based on Bi-GDM fitting and the K-means clustering algorithm can describe heartbeat intervals without training a network model, particularly when the radar sensor detecting vital signs has a high-frequency resolution and high sample quality. BBIs variation sequence curves based on the proposed method are consistent with RR time sequences detected by ECG. When the sampling object is unknown, and there is no reference signal for synchronous sampling, the proposed BBIs detection method can describe heartbeat intervals and heartbeats in different cycle phases.

The comparison between existing studies and this study is shown in Table 5. The proposed method's performance with cardiac sequence segmentation is better than those of other existing methods. The heartbeat cycle is segmented into subsets that are associated with essential points in ECG. The existing techniques listed in Table 5 segment cardiac sequences using heart sound stage division as a reference and use envelopes extracted from filtered radar phase signals as features. The logistic regression-HSMM-based heart sound segmentation method [23] was used in [12] to segment the heart sound stage, and the

Table 5

Comparison with the state-of-the-art approaches.

Paper	Model	Input features	Target	Performance
[12]	HSMM	Envelopes	Heart sound	$F_{1,S1} = 93.15\% \pm 10.75\%$
[24]	Bi-LSTM	Envelopes	Heart sound	$F_{1,S1} = 95.8\%$; Acc. = 93.4%
Proposed	RNNs (Bi-LSTM; LSTM; GRU)	Three clustered waves based on Bi-GDM fitting and K-means	ECG segments	$F_{1,RT} = 92.46\%$
				$\pm 6.53\%$;
				$F_{1,TR} = 97.14\%$
				$\pm 2.15\%$;
				Acc. = $95.88\% \pm 3.19\%$
				$F_{1,QR} = 76.78\%$
				$\pm 15.94\%$;
				$F_{1,RS} = 78.08\%$
				$\pm 16.45\%$;
				$F_{1,ST} = 91.41\%$
				$\pm 7.31\%$;
				$F_{1,TQ} = 96.86\%$
				$\pm 2.39\%$;
				Acc. = $93.86\% \pm 4.49\%$

$F_{1,S1}$ scores were approximately 93%. Reference [24] evaluated different LSTM architectures and concluded that Bi-LSTM has the best performance in cardiac sequence segmentation. The valuation result of heart sound segmentation is $F_{1,S1} = 95.8\%$, and Acc. = 93.4% for both heart sound stages. For the same training features and radar system, $F_{1,S1}$ of [24] using Bi-LSTM is higher than the score in [12] using the HSMM model. In the proposed method, three clustered waves based on Bi-GDM fitting and K-means are pre-processed and selected as model features. Three RNN models are examined in the heartbeat sequences segmentation. The classification results of sequence “R-T-R” using Bi-LSTM exhibit higher scores and accuracy, with $F_{1,RT} = 92.46\%$, $F_{1,TR} = 97.14\%$ and Acc. = 95.88%. For the sequence “Q-R-S-T-Q” using Bi-LSTM, the classified results are $F_{1,QR} = 76.78\%$, $F_{1,RS} = 78.08\%$, $F_{1,ST} = 91.41\%$, $F_{1,TQ} = 96.86\%$ and Acc. = 93.86%. Thus, the proposed method achieves the better performance in cardiac sequence detection than that of the state-of-the-art approaches. Moreover, the attempt of classifying the cardiac radar sequences using the ECG sequences directly is proposed for the first time.

5. Conclusions

The model RBHMM is proposed based on heartbeat component distribution in the time and frequency domains using radar sensing. Then, the cardiac cycle detection method based on RBHMM is analysed. The signals of heartbeat high-frequency components include two distinct parts related to ventricular systolic and diastolic variation. These central frequencies of the two parts can be separated by Bi-GDM fitting. Compared with other frequency ranges, the frequency range of 3–15 Hz is the most suitable range to extract a heartbeat signal. The extracted signals based on detected central frequencies can be clustered into three classes using K-means, which correspond to R-peaks, T-peaks and the T-wave end of ECG. The cardiac cycle detection methods to extract BBIs and cardiac cycle segmentation were evaluated. BBIs extraction using low-pass filtering can roughly detect heartbeat cycles. The method based on Bi-RNN has a good performance in TQ and TR sequence segmentation. The accuracy values to segment and classify sequences into R-T-R and “Q-R-S-T-Q” are $95.88\% \pm 3.19\%$ and $93.86\% \pm 4.49\%$, respectively. Thus, heart disease can be diagnosed more accurately using segmented radar signal sequences.

CRedit authorship contribution statement

Shanling Ji: Conceptualization, Methodology, Software, Validation, Writing – original draft, Visualization. **Zhisheng Zhang:** Resources, Supervision, Project administration, Funding acquisition. **Zhijie Xia:** Writing – review & editing, Validation. **Haiping Wen:** Writing – review & editing, Project administration, Formal analysis, Investigation, Funding acquisition. **Jianxiong Zhu:** Writing – review & editing, Visualization. **Kunkun Zhao:** Writing – review & editing, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] A. Singh, S.U. Rehman, S. Yongchareon, P. Chong, Multi-resident Non-Contact Vital Sign Monitoring using Radar: A Review, *IEEE Sens. J.* 21 (4) (2021) 4061–4084, <https://doi.org/10.1109/JSEN.2020.3036039>.
- [2] A. Al-Najji, K. Gibson, S.H. Lee, J. Chahl, “Monitoring of Cardiorespiratory Signal: Principles of Remote Measurements and Review of Methods, *IEEE Access* (2017) 15776–15790, <https://doi.org/10.1109/ACCESS.2017.2735419>.
- [3] E. Antolinos, F. García-Rial, C. Hernández, D. Montesano, J.I. Godino-Llorente, J. Grajal, Cardiopulmonary Activity Monitoring Using Millimeter Wave Radars, *Remote Sens.* 12 (14) (2020) 2265.
- [4] S. Dong, Y.i. Zhang, C. Ma, C. Zhu, Z. Gu, Q. Lv, B. Zhang, C. Li, L. Ran, Doppler Cardiogram: A Remote Detection of Human Heart Activities, *IEEE Trans. Microw. Theory Tech.* 68 (3) (2020) 1132–1141.
- [5] M. Mercuri, Y. Lu, S. Polito, F. Wieringa, Y.-H. Liu, A.-J. van der Veen, C. Van Hoof, T. Torfs, Enabling Robust Radar-Based Localization and Vital Signs Monitoring in Multipath Propagation Environments, *IEEE Trans. Biomed. Eng.* 68 (11) (2021) 3228–3240.
- [6] S. Yoo, S. Ahmed, S. Kang, D. Hwang, J. Lee, J. Son, S.H. Cho, Radar Recorded Child Vital Sign Public Dataset and Deep Learning-Based Age Group Classification Framework for Vehicular Application, *Sensors* 21 (7) (2021) 2412.
- [7] C.M. Astley, et al., Remote cardiac rehabilitation services and the digital divide: implications for elderly populations during the COVID19 pandemic, *Eur. J. Cardiovasc. Nurs.* 20 (6) (2021) 521–523, <https://doi.org/10.1093/eurjcn/zvab034>.
- [8] S.H. Oh, S. Lee, S.M. Kim, J.H. Jeong, Development of a heart rate detection algorithm using a non-contact doppler radar via signal elimination, *Biomed. Signal Process. Control* 64 (2020) 102314, <https://doi.org/10.1016/j.bspc.2020.102314>.
- [9] F. Michler et al., “A Clinically Evaluated Interferometric Continuous-Wave Radar System for the Contactless Measurement of Human Vital Parameters,” (in eng), *Sensors (Basel)*, vol. 19, no. 11, May 31 2019, doi: 10.3390/s19112492.
- [10] M. Nosrati, N. Tavassolian, Accurate Doppler Radar-Based Cardiopulmonary Sensing Using Chest-Wall Acceleration, *IEEE J. Electromagn. RF Microwaves Med. Biol.* 3 (1) (2018) 41–47, <https://doi.org/10.1109/JERM.2018.2879452>.
- [11] W. Xia, Y. Li, S. Dong, Radar-Based High-Accuracy Cardiac Activity Sensing, *IEEE Trans. Instrum. Meas.* 70 (2021) 1–13, <https://doi.org/10.1109/TIM.2021.3050827>.
- [12] C. Will, K. Shi, S. Schellenberger, T. Steigleder, F. Michler, J. Fuchs, R. Weigel, C. Ostgathe, A. Koelpin, Radar-Based Heart Sound Detection, *Sci. Rep.* 8 (1) (2018), <https://doi.org/10.1038/s41598-018-29984-5>.
- [13] L. Sörnmo, P. Laguna, *Electrocardiogram (ECG) Signal Processing*, Wiley Encycl. Biomedical Eng. 2 (2006) 1298–1313, <https://doi.org/10.1002/9780471740360.ebs1482>.
- [14] E.H. Locati, G. Bagliani, Heart Rate Dependency of QT-Interval in Congenital and Acquired Prolonged Ventricular Repolarization: Long-Term Analysis by Holter Monitoring, *Adv. Noninvasive Electrocardiographic Monitor. Techniques* 229 (2000), https://doi.org/10.1007/978-94-011-4090-4_14.
- [15] N. Naseer, H. Nazeer, Classification of normal and abnormal ECG signals based on their PQRS intervals, in: 2017 International Conference on Mechanical, System

- and Control Engineering (ICMSC), 2017, pp. 388–391, doi: 10.1109/ICMSC.2017.7959507.
- [16] Y. Xiong, Z. Peng, C. Gu, S. Li, D. Wang, W. Zhang, Differential Enhancement Method for Robust and Accurate Heart Rate Monitoring via Microwave Vital Sign Sensing, *IEEE Trans. Instrum. Meas.* 69 (9) (2020) 7108–7118.
 - [17] R. Yu, D.W. Bliss, Remote Sensing for Vital Information Based on Spectral-Domain Harmonic Signatures, *IEEE Trans. Aerosp. Electron. Syst.* 55 (6) (2019) 3454–3465, <https://doi.org/10.1109/TAES.2019.2917489>.
 - [18] Y. Ding, X. Yu, C. Lei, Y. Sun, X. Xu, J. Zhang, A Novel Real-Time Human Heart Rate Estimation Method for Noncontact Vital Sign Radar Detection, *IEEE Access* 8 (2020) 88689–88699.
 - [19] N.D. Chinh, et al., Short time cardio-vascular pulses estimation for dengue fever screening via continuous-wave Doppler radar using empirical mode decomposition and continuous wavelet transform, 2021/03/01/, *Biomed. Sign. Process. Control* 65 (2021), <https://doi.org/10.1016/j.bspc.2020.102361>.
 - [20] X. Hu, T. Jin, Short-Range Vital Signs Sensing Based on EEMD and CWT Using IR-UWB Radar, *Sensors* 16 (12) (2016) 2025, <https://doi.org/10.3390/s16122025>.
 - [21] Y. Wang, W. Wang, M. Zhou, A. Ren, Z. Tian, Remote Monitoring of Human Vital Signs Based on 77-GHz mm-Wave FMCW Radar, *Sensors* 20 (10) (2020) 2999, <https://doi.org/10.3390/s20102999>.
 - [22] M. Nosrati, N. Tavassolian, High-Accuracy Heart Rate Variability Monitoring Using Doppler Radar Based on Gaussian Pulse Train Modeling and FTFR Algorithm, *IEEE Trans. Microw. Theory Tech.* 66 (1) (2017) 556–567, <https://doi.org/10.1109/TMTT.2017.2721407>.
 - [23] D.B. Springer, L. Tarassenko, G.D. Clifford, Logistic Regression-HSMM-Based Heart Sound Segmentation, *IEEE Trans. Biomed. Eng.* 63 (4) (2016) 822–832, <https://doi.org/10.1109/TBME.2015.2475278>.
 - [24] Kilin et al., “Segmentation of Radar-Recorded Heart Sound Signals Using Bidirectional LSTM Networks,” 2019 41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), vol. 2019, pp. 6677–6680, 2019, doi: 10.1109/EMBC.2019.8857863.
 - [25] Y. Chen, J. Lv, Y. Sun, B. Jia, Heart sound segmentation via Duration Long-Short Term Memory neural network, 2020/10/01/, *Appl. Soft Comput.* 95 (2020) 106540, <https://doi.org/10.1016/j.asoc.2020.106540>.
 - [26] E. Messner, M. Zöhrer, F. Pernkopf, Heart Sound Segmentation—An Event Detection Approach Using Deep Recurrent Neural Networks, *IEEE Trans. Biomed. Eng.* 65 (9) (2018) 1964–1974, <https://doi.org/10.1109/TBME.2018.2843258>.
 - [27] J. Wang, X. Wang, L. Chen, J. Huangfu, C. Li, L. Ran, Noncontact Distance and Amplitude-Independent Vibration Measurement Based on an Extended DACM Algorithm, *IEEE Trans. Instrum. Meas.* 63 (1) (2014) 145–153, <https://doi.org/10.1109/TIM.2013.2277530>.
 - [28] B. Park, O. Boric-Lubecke, V.M. Lubecke, Arc tangent Demodulation With DC Offset Compensation in Quadrature Doppler Radar Receiver Systems, *IEEE Trans. Microw. Theory Tech.* 55 (5) (2007) 1073–1079, <https://doi.org/10.1109/TMTT.2007.895653>.
 - [29] V.L. Petrović, M.M. Janković, A.V. Lupšić, V.R. Mihajlović, J.S. Popović-Božović, High-Accuracy Real-Time Monitoring of Heart Rate Variability Using 24 GHz Continuous-Wave Doppler Radar, *IEEE Access* 7 (2019) 74721–74733, <https://doi.org/10.1109/ACCESS.2019.2921240>.
 - [30] G. Tse, et al., “T(peak)-T(end), T(peak)-T(end)/QT ratio and T(peak)-T(end) dispersion for risk stratification in Brugada Syndrome: A systematic review and meta-analysis,” (in eng), *J. Arrhythm.* 34 (6) (Dec 2018) 587–597, <https://doi.org/10.1002/joa3.12118>.
 - [31] P.E. McSharry, G.D. Clifford, L. Tarassenko, L.A. Smith, A Dynamical Model for Generating Synthetic Electrocardiogram Signals, *IEEE Trans. Biomed. Eng.* 50 (3) (2003) 289–294.
 - [32] M.S. Manikandan, K.P. Soman, A novel method for detecting R-peaks in electrocardiogram (ECG) signal, *Biomed. Signal Process. Control* 7 (2) (2012) 118–128, <https://doi.org/10.1016/j.bspc.2011.03.004>.
 - [33] S. Schellenberger, K. Shi, T. Steigleder, A. Malessa, F. Michler, L. Hameyer, N. Neumann, F. Lurz, R. Weigel, C. Ostgathe, A. Koelpin, A dataset of clinically recorded radar vital signs with synchronised reference sensor signals, *Sci. Data* 7 (1) (2020), <https://doi.org/10.1038/s41597-020-00629-5>.
 - [34] Sedghamiz, BioSigKit: A Matlab Toolbox and Interface for Analysis of BioSignals, *J. Open Source Softw.* 3(30) (2018). doi: 10.21105/joss.00671.